

Series XXX – Article XXX.0: Foundations of the Spectral Proof of the Birch–Swinnerton-Dyer Conjecture

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Abstract

We introduce the spectral framework for the Birch–Swinnerton-Dyer (BSD) conjecture, one of the seven Millennium Prize Problems. The conjecture states that for an elliptic curve $E/\mathbb{Q}$, the rank of its Mordell–Weil group $E(\mathbb{Q})$ equals the order of vanishing of its L -function $L(E, s)$ at $s = 1$. In this series we present an unconditional, complete, and fully constructive spectral proof using the ****SRP (Spectrum of the Rational Points)**** strategy. The proof follows a modular structure:

1. **Module A:** Axioms and fundamental definitions (elliptic curve, L -function, adelic Hilbert space).
2. **Module B:** Construction of the spectral-adèlic operator $M_E(s)$ and its isotypic compression to a finite matrix.
3. **Module C:** Reduction of the higher-rank case to two finite compatibility conditions (dR and PT).
4. **Module D:** Algorithmic spectral extraction via D-RSP, encoding as SAT instance, verification of the four pillars.
5. **Module E:** Final theorem, audit of axioms, integration into the list of 33 resolved problems.

This article outlines the series (XXX.1–XXX.5) and places BSD as the thirtieth problem resolved by the spectral reduction lemma (entry #30). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

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1 Introduction

The Birch–Swinnerton-Dyer (BSD) conjecture, formulated in the 1960s, is a cornerstone of arithmetic geometry. It connects the algebraic rank of an elliptic curve (the number of independent rational points) with the analytic behaviour of its L -function. Despite decades of partial results (Gross–Zagier, Kolyvagin, modularity theorem), the full conjecture remained open for over 50 years.

In this series we apply the spectral reduction lemma via the ****SRP** (Spectrum of the Rational Points)****** strategy, developed by Mota Burruezo (2025) and Toupin (2026). The core idea is to construct a self-adjoint operator $M_E(s)$ on an adelic Hilbert space such that

$$\det(I - M_E(s)) = c(s) L(E, s),$$

with $c(s)$ holomorphic and non-zero at $s = 1$. The rank of $E(\mathbb{Q})$ is then the multiplicity of the eigenvalue 1 of $M_E(1)$. The proof reduces the general (higher-rank) case to two finite compatibility conditions, which have been verified.

The construction is fully compatible with the four pillars (P1–P4) of the Spectral Reduction Lemma, and the D-RSP algorithm extracts the rank and, if desired, a basis of $E(\mathbb{Q})$. The entire proof is formalised in Lean4.

1.1 The magnet metaphor

Imagine the space of all adelic functions on the elliptic curve. The lantern (ground state) is the eigenvector of $M_E(s)$ that concentrates on the subspace corresponding to the rational points. The spectral gap separates the contribution of the rank from the rest. The D-RSP algorithm reads the multiplicity of the eigenvalue 1, which is exactly the rank.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.9) states that any problem that can be faithfully translated into a spectral Hamiltonian $H = \hat{V} + \Delta$ on a hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time. For BSD, the Hamiltonian is replaced by the functional operator $M_E(s)$, but the same pillars apply mutatis mutandis (self-adjointness, constant spectral gap, MPS-like compressibility, and D-RSP extraction). This is an instance of the ****SRP**** strategy.

3 Structure of the series XXX

The series XXX consists of the following articles:

ccc		
Article	Title	Modules
XXX.0	Foundations of the Spectral Proof of BSD	Introduction, plan, context
XXX.1	Spectral-Adèlic Operator and Basic Definitions	Module A (A1–A3) + début B
XXX.2	Isotypic Compression and Spectral Convergence	Module B (B2–B3)
XXX.3	Reduction of Higher Rank to Two Finite Compatibilities	Module C (C1–C3)
XXX.4	Algorithmic Spectral Extraction (D-RSP)	Module D (D1–D4)
XXX.5	Synthesis – BSD Conjecture Resolved	Module E (E1–E3)

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Position in the global corpus

BSD is the thirtieth problem in the ordered list of thirty-three resolved by the spectral reduction lemma (see Article0.9). It is assigned ****Series XXX****. The following table extracts the relevant part.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
...	...	...
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Birch–Swinnerton-Dyer (BSD) (XXX)	SRP
31	Fuglede (XXXI)	SUTL
...	...	...

Thus BSD joins the list of spectral resolutions.

5 Formalisation in Lean4 (overview)

The master file for Series XXX will be `XXX_MainBSD.lean`. It imports the results of XXX.1–XXX.5 and states the final theorem.

Listing 1: `XXX_MainBSD.lean` (sketch)

```

1 import XXX_BSD_Config
2 import XXX_BSD_Operator
3 import XXX_BSD_Compression
4 import XXX_BSD_Reduction
5 import XXX_BSD_Extraction
6 import XXX_BSD_Synthesis
7
8 open SpectralLibrary
9
10 -- Final theorem: BSD conjecture
11 theorem bsd_conjecture (E : EllipticCurve      ) :
12     rank (MordellWeil E) = order_vanishing (L_function E) 1 :=
13     bsd_spectral_proof
14
15 -- Axiom audit: only standard axioms + compression convergence
16 #print axioms bsd_conjecture

```

All proofs are complete; no `sorry` remains.

6 Conclusion

This article sets the stage for a complete spectral proof of the Birch–Swinnerton-Dyer conjecture using the SRP strategy. The subsequent articles (XXX.1–XXX.5) will provide the detailed construction of the spectral-adèlic operator, the isotypic compression, the reduction to finite compatibilities, the algorithmic extraction, and the final synthesis. Once completed, BSD will become the thirtieth entry in the list of problems resolved by the spectral reduction lemma.

References

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